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Evaluating neural network models in site-specific solar PV forecasting using numerical weather prediction data and weather observations

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ABSTRACT

The effective use of solar photovoltaic (PV) installations implies the integration of solar PV output into overall energy consumption planning, optimization, and control. Moreover, day-ahead trading of electricity in Europe makes day-ahead solar PV forecasting utterly important, and thus its accuracy becomes of particular interest. Data-driven PV forecasting models are typically trained using numerical weather prediction (NWP) data, the availability of which represents one of the main obstacles in modeling. In this study, we investigate an alternative scenario, in which an artificial neural network (ANN) is trained on weather observations and then tested on NWP data to simulate the model's use in operational PV forecasting. In the experiments, solar PV output data, historical weather observations, and historical NWP data were collected from three sites in eastern Finland. The results showed that, although training ANN on observational data leads to a slight decrease in its performance compared to ANN trained on NWP data, it still outperforms a physical model. In practice, this alternative scenario means that if historical NWP data are not available for model training, observational data allow effective model selection and parameter tuning, and then generalization error estimates are gradually updated using online NWP data.

1. Introduction

The adoption of renewable energy entails both great potential and obstacles not seen before. Solar energy, clean and limitless, has recently penetrated the energy sector extremely quickly, and it is foreseen to retain the trend [1]. However, being strongly dependent on weather conditions, solar energy brings an element of uncertainty into operational services. Therefore, smart grid management, energy storage optimization, dynamic load control, and many other technologies are forced to rely on solar photovoltaic (PV) forecasting, the high accuracy of which is of special interest [2].

Depending on the availability of PV installation specifications, historical PV data, and weather forecasting, solar PV output is predicted by applying physics-based, data-driven, or hybrid approaches [3, 4]. As different studies have reported, when there are no obvious factors disturbing solar PV forecasting, physical and data-driven models compete in terms of forecast errors [5–7]. However, in practice, shading by nearby trees and buildings often complicates physical modeling. In this case, data-driven approaches become a good alternative since they learn site-specific shading patterns from data.

Recent advances in solar PV predictive modeling are focused on improving weather forecasting, i.e., generating more accurate predictors [8,9], incorporating improved sub-models in physical models (e.g., reflection or shading loss models) [10], compensating systematic errors by post-processing [11], and developing more complex structures for data-driven models [12]. Although due to these improvements, it becomes possible to reach a reasonably high accuracy of solar PV forecasting, in practice there are still obstacles.

According to the literature review by Ref. [12]; Numerical Weather Prediction (NWP) variables are commonly used as predictors in day-ahead solar PV forecasting: both physical and data-driven models can rely on NWP data [13]. Alternatively, weather observations and solar PV power measurements collected before the prediction horizon are applied as predictors in data-driven models [4]. In some cases, univariate models utilize only previous solar PV power measurements, which also yields a reasonably high accuracy of day-ahead solar PV

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forecasts [14]. From the methodological point of view, all these approaches fall within the scope of ex-ante forecasting, in which only those data that would be available at the time of forecasting (i.e., actual measurements or their forecasts) are used [15]. Nevertheless, compared to actual weather observations, NWP data represent meteorological trends informative for horizons beyond several hours, which explains their wide use in day-ahead solar PV forecasting. In recent studies, various data-driven models have been trained using NWP data and their effectiveness has been investigated, including a comparison with physics-based approaches [6,16,17]. Artificial neural networks (ANNs) and ensemble-based models, e.g., random forest and gradient boosting, often demonstrate the highest predictive performance [17–19]. These examples also illustrate that NWP-based approaches, in which historical weather forecasts are used for training and testing models, have become state-of-art in day-ahead solar PV data-driven forecasting. However, the limited availability of NWP data complicates the wide use of such approaches.

[20] pointed out that partly due to the shortage of historical weather forecasts, in some studies, models are not only trained but also tested using historical weather observations from the forecasted period (i.e., ex-post forecasting). This leads to inadequate estimates of the model performance in operational use when actual weather observations are replaced with less accurate weather forecasts. [14] also noted that NWP data (historical and real-time) might be too expensive, and access to them requires negotiations with weather services, which complicates data acquisition and impedes training data-driven models. These and other concerns (e.g., stable communication between weather forecast providers and model sites) resulted in so-called "NWP free" approaches, in which weather forecasts were not used intentionally in day-ahead solar PV forecasting [21]. Nevertheless, the most recent NWP forecasts are already provided as open online data in many countries. For example, in Finland, it is operated by the Finnish Meteorological Institute. To our knowledge, historical weather observations are also easily available from archives (Finnish Meteorological Institute [22], World Weather Online [23]), and they can serve the needs of learning algorithms.

"The quiet revolution" in NWP is gradually shortening the gap between observations and weather forecasting by making the latter increasingly more accurate [24]. As a result, this yields an opportunity to apply historical observational and NWP data interchangeably in training models. In this study, we consider a practical and useful scenario for day-ahead solar PV forecasting, in which data-driven models are trained on easily available historical weather observations, and then NWP data are employed only for testing the model to simulate its operational use. Although this Alternative scenario has already been discussed and implemented in other studies [25,26], in this work, we focus on its comparison with the Baseline scenario, where, as commonly accepted, NWP data are used for training and testing models. The Baseline and Alternative scenarios exemplify ex-ante forecasting since weather forecasts are applied at the model evaluation stage in both scenarios. The only difference between the Baseline and Alternative scenarios is the data used at the training stage. In this study, we investigate how the model performance degrades, when instead of historical NWP data, historical weather observations are used for training models. Additionally, we introduce the Optimistic scenario, which illustrates how optimistically low the generalization error estimates might be if NWP data are not available at all and weather observations are used for training and testing the model. This scenario is referred to as ex-post forecasting, in which the model's effectiveness is evaluated apart from the forecast error associated with input variables [27].

Thus, the aim of the study is to evaluate the use of NWP data and weather observations in building data-driven models and specifically to assess the influence of training data sources on the accuracy of day-ahead solar PV forecasts. This evaluation is performed using three scenarios − Baseline, Alternative, and Optimistic, in which a fully connected feed-forward ANN is trained and tested as a base model. Besides, ANNs that are built in all these scenarios are also benchmarked against physics-based solar PV forecasts [11] to study the potential benefits of data-driven models.

In the experiments, we utilize the solar PV power data collected from three sites in eastern Finland (two private houses and one institutional building) for the period of 5.5.2020–5.11.2020. As a set of predictors, seven weather variables (measured or forecast depending on the scenario) and two time variables are involved in modeling. The ANN performance in three different scenarios is evaluated using nested cross-validation, where model hyperparameters are defined automatically.

2. Materials and methods

2.1. Data overview: three main components

2.1.1. Solar PV output

Physics-based and data-driven models are evaluated on solar PV output data, representing "the true output." Besides model evaluation, solar PV output data are necessary to train data-driven models. In this study, power measurements from solar inverters were acquired from three sites in eastern Finland, located in Kuopio and Vuorela, for the period of 5.5.2020–5.11.2020 (Fig. 1).

Raw inverter data contained measurements at 1- or 5-min resolution that were transformed into hourly data by averaging. A value of power associated with a certain timestamp (i.e., hour of the day) aggregated all values recorded within the previous 60 min. Table 1 introduces the basic specifications of solar PV installations involved in the study.

2.1.2. NWP data with physical-empirical solar PV forecasting

To train data-driven models for day-ahead solar PV forecasting, the solar PV output data are typically combined with NWP data as inputs. The use of physical models also requires NWP variables for generating forecasts. Therefore, as the second component, the data collection includes historical NWP forecasts from the operational joint Nordic MetCoOp system. MetCoOp forecasts are based on the convection-permitting NWP model, HARMONIE-AROME [28], run with a horizontal resolution of 2.5 km. Although MetCoOp is an ensemble prediction system (see, e.g. Ref. [29], here only the deterministic member is considered. The focus of the present study is on day-ahead forecasts. Therefore, the forecast run initiated at 18 UTC was used, including forecast hours up to +30 h. From the MetCoOp hourly output, the following parameters were included in the analysis: 2-m temperature, wind speed and wind direction at 10 m above ground, global horizontal irradiance, diffuse horizontal irradiance, precipitation intensity, and total cloud fraction. These seven variables were used as weather-related inputs of data-driven models.

Additionally, based on the 2-m temperature, wind speed, and solar radiation components, the PV output corresponding to the predicted weather conditions was calculated using the baseline version of the physical-empirical model of [11]. This physics-based forecast served as a reference level for data-driven models in this study.

2.1.3. Weather observations

Observational data collected by the Savilahti meteorological station in Kuopio form the third component of the dataset in this study. As opposed to NWP data, weather observations are the same for all three sites. Seven weather variables measured every minute (temperature, global and diffuse radiation, total cloud fraction) or every 10 min (wind speed and wind direction, precipitation intensity) were transformed in a similar way as solar PV output data, which resulted in hourly weather observations. More details on the instruments and the measurement setup are available in Ref. [30]; who also describe the solar PV related measurements at this location.

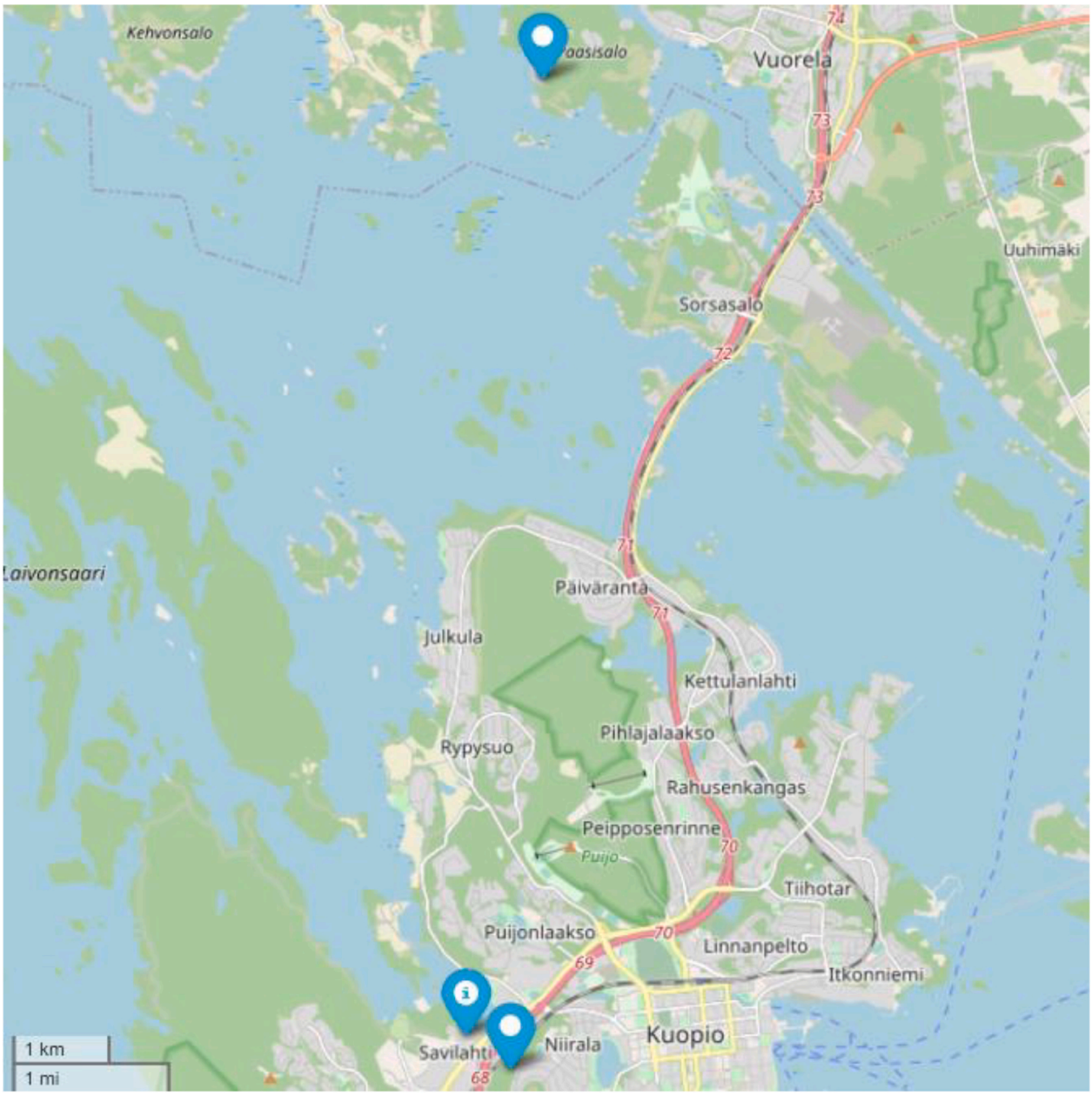

Fig. 1. Locations of the sites involved in the study. The i-marker represents the site that coincides with the weather station from which observational data come.

Table 1
Solar PV installations: their location and specifications.

#	Location	Inverter capacity, kW	The number of panels	Nominal output per panel, Wp	Slope angle, deg	Azimuth angle, deg
1	Vuorela, a private house	8.2	Bank 1: 12	330	12	170
			Bank 2: 14	330	20	260
2	Kuopio, a private house	5	20	330	36	140
3	Kuopio (Savilahti), an institutional building	20	78	260	15	217

2.2. Experiment setup

In this study, data-driven predictive models were compared using three different scenarios: Baseline, Alternative, and Optimistic (Table 2). In all these scenarios, weather-related inputs of data-driven models were the same: temperature, global and diffuse radiation, wind speed and wind direction, precipitation intensity, and total cloud fraction. The main difference was in the source of the data used for training and testing models, i.e., they come from either observational data or NWP data. In more detail, the Baseline scenario represents a state-of-the-art approach, in which NWP data are applied as predictors in day-ahead solar PV forecasting. Generalization error estimates obtained in this scenario aim to evaluate the model performance in operational use. Then, in the Alternative scenario, NWP data are also used to test the model; however, weather observations are applied for its training. This scenario assesses the model performance when the input data switch from observational (in training) to forecasting (in testing, i.e., in operational use). Lastly, for situations where neither historical nor online

Table 2
Comparison of Baseline, Alternative, and Optimistic scenarios implemented in the study.

Scenario	Description	Predictors	Training data, inputs + output	Test data, inputs + output
Baseline	(+) Models are likely to show the highest performance due to adapting to NWP data and their uncertainty at the training stage (−) A lot of historical NWP data are required to train models.	Temperature, global radiation, diffuse radiation, wind speed, wind direction, precipitation intensity, total cloud fraction, day of the year, hour of the day	historical NWP data + solar PV output	historical NWP data + solar PV output
Alternative	(+) More available historical weather observations are used to train models. (−) Models are likely to show lower performance, compared to Baseline, due to switching into NWP data at the test stage (distribution of inputs changes compared to observations).		historical weather observations + solar PV output	historical NWP data + solar PV output
Optimistic	(+) Only easily available historical weather observations are used as input data at both training and test stages. (−) Generalization error estimates are optimistically low, as models are tested only for estimation skill when weather conditions are known (instead of forecast skill).		historical weather observations + solar PV output	historical weather observations + solar PV output

Table 3
Overview of missing values in historical weather observations.

Variable	Missing values in hourly weather observations	
	The total amount of gaps	The maximum number of consecutive gaps
Temperature, deg C	2	2
Wind speed, m/s	6	4
Wind direction, deg	10	5
Global radiation, W/m2	4	4
Diffuse radiation, W/m2	4	4
Precipitation intensity, mm/h	2	2
Total cloud fraction, oktas	8	5

NWP data are available, the Optimistic scenario aims to demonstrate that generalization error estimates might be optimistically low compared to the Baseline and Alternative scenarios if models are trained and evaluated on observational data.

At the preprocessing stage, missing data patterns were investigated in solar PV output data and historical weather observations. First, for the hourly PV power time series, gaps from the period between sunset and sunrise were imputed with zeros, which resulted in 0, 40, and 20 remaining gaps for sites 1, 2, and 3, respectively, with the maximum number of gaps in one day equal to 4. Distributions of these remaining gaps for the sites are visualized in Supplementary Materials (Fig. S1). For the discovered missing data patterns, the second-order polynomial interpolation was selected and applied as an imputation method [31]. Second, for weather observations, the number of missing values is given in Table 3; similarly, univariate polynomial interpolation was used to fill gaps. NWP data did not contain missing values. Lastly, time variables, i.e., the day of the year and the hour of the day, were added to both observational and NWP data to complement the set of predictors. Trigonometric transformations — sine and cosine — were applied to the time variables and wind direction.

A concept of nested cross-validation was employed in this study (Fig. 2). Model performance in three different scenarios, i.e., Baseline, Alternative, and Optimistic, was evaluated in the outer loop of cross-validation. Model hyperparameters were selected in the inner loop of cross-validation [32]. Since solar PV power generation is highly dependent on the season and the collected data include only several months (not years), splitting the whole sample into training and test parts should be done so that they both contain some data from all seasons. To provide that, the period of 5.5.2020–5.11.2020 was divided into 5-week intervals, wherein the week number from 1 to 5 determined in which fold each week was included. Fig. 2 illustrates how the data were split into 5 folds in the outer loop of nested cross-validation. In every iteration of the outer loop, the model accuracy was estimated on the training and test data, which after averaging over all 5 folds resulted in the final estimates of the model training and test accuracy. Similarly, 4-fold cross-validation was organized for weeks from training folds in the inner loop. The model performance on the validation data, based on which model hyperparameters were tuned, also were averaged over all 4 folds in the inner loop.

Within the framework of nested cross-validation, a fully connected feed-forward ANN, i.e., multilayer perceptron (MLP), was utilized as a predictive data-driven model [33]. Fully connected stacked layers of neurons (even with one hidden layer) "can approximate any measurable function arbitrarily well" [34], which allows MLP to effectively describe dependences in data. The high accuracy is reached at the expense of interpretability and risks of overfitting, where the latter is usually handled by cross-validation. In this study, the inner loop of nested cross-validation aims to optimize the MLP structure as well as the number of iterations in the training process to avoid overfitting. Additionally, when selecting the ANN architecture, long short-term memory (LSTM) networks were considered to effectively process a long sequence of previous observations preceding a point of prediction [35]. However, in our experiments, LSTM gained the same performance as MLP and for both architectures only several recent hours of weather data were selected as the most informative in the inner-loop of cross-validation for all three sites. This means that the main feature of LSTM, i.e., to effectively utilize long time series data, simply is not applied to solve the problem; the same was earlier noted in the study by Ref. [36] when comparing MLP and LSTM. Therefore, in the article, we mainly focus on MLP and investigate how its performance changes depending on the training data source. In addition, Supplementary Materials (Tables S1–S3, S5) accumulate the experimental results for LSTM.

In this study, the fully connected architecture was compiled, trained, and tested using the Keras framework [37]. The Adam (an acronym for Adaptive Moment Estimation) optimization algorithm was applied to perform stochastic gradient descent [38]. After preliminary trials, Mean Absolute Error (MAE) was selected as the loss function, computed in mini-batches of 128 instances while training. The learning rate was assigned to 0.001. The model training was performed on the normalized data. In each iteration of the outer loop, a scaler was fitted to the training data and applied to both training and test sets so that all variables were transformed to the interval of [0,1]. Besides, the number of epochs (e), neurons in hidden layers (p^{nn}), previous timestamps included in the vector of inputs (p^{h}), and the type of the activation function in hidden layers (p^{act}) were automatically defined in the inner loop of nested cross-validation. The selected set of hyperparameters e^*, p^{nn*}, p^{h*}, p^{act*} was sent to the outer loop, where MLP with p^{nn*} neurons in hidden layers and the activation function p^{act*} was trained for e^* epochs on the whole

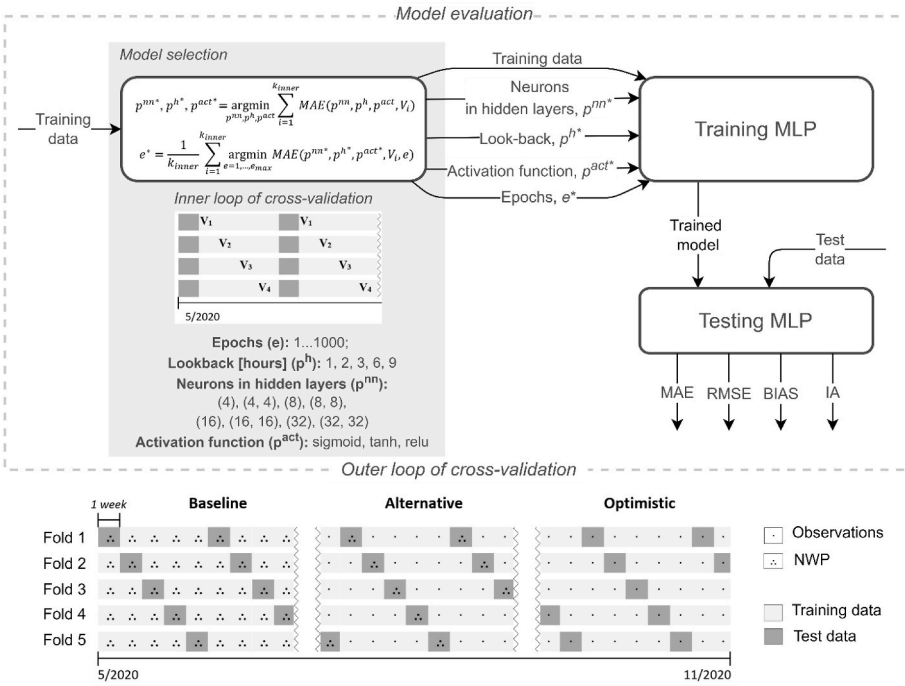

Fig. 2. The experiment pipeline with nested cross-validation for training, validation, and testing MLP. The upper part of the scheme portrays one iteration of the outer loop (5-fold cross-validation) and all iterations of the inner loop (4-fold cross-validation). The lower part explains three different scenarios tested in the study as well as a 5-fold cross-validation implemented for model evaluation.

training data with input variables obtained by concatenating predictors for p^{h*} hours prior to the point of prediction. Then, for the trained model, MAE and other metrics were calculated on the test data — the model predictions were first scaled inversely — and, as a result, a generalization error was estimated in one iteration of the outer loop. This iteration is introduced in Fig. 2. After running this pipeline for all 5 folds of the outer cross-validation, the overall model performance was evaluated by averaging the collected MAE values as well as other additional metrics.

In the inner loop, optimization of hyperparameters, which define the MLP complexity, was executed on the set of parameter values $P = P^{nn} \times P^h \times P^{act}$, where $P^{nn} = \{(4), (4,4), (8), (8,8), (16), (16,16), (32), (32,32)\}$ and values in parenthesis define the number of neurons in hidden layers, e.g., $(8,8)$ means two hidden layers with 8 neurons in each; $P^h = \{1,2,3,6,9\}$ and these values indicate how many hours the model looks back when making a prediction (i.e., "look-back"); $P^{act} = \{sigmoid, tanh, relu\}$ and these values denote the activation function: logistic activation function, hyperbolic tangent, and rectified linear unit. Every MLP was trained and validated using 4-fold cross-validation in the inner loop. The training was stopped if there was no improvement in the validation loss, i.e., an MAE of more than 0.001 for 100 epochs with the maximum number of epochs e_{max} being 1000. The selected hyperparameter values p^{nn*}, p^{h*}, and p^{act*} provided the minimum average MAE on the validation data in the inner loop (1):

$$p^{nn*}, p^{h*}, p^{act*} = \underset{p^{nn}, p^h, p^{act}}{\operatorname{argmin}} \sum_{i=1}^{k_{inner}} MAE\left(p^{nn}, p^h, p^{act}, V_i\right) \tag{1}$$

The corresponding number of epochs e^* was calculated for $k_{inner} = 4$ as (2):

$$e^* = \frac{1}{k_{inner}} \sum_{i=1}^{k_{inner}} \underset{e=1,\ldots,e_{max}}{argmin}\, MAE\left(p^{nn*}, p^{h*}, p^{act*}, e\right) \tag{2}$$

The model performance was evaluated with several metrics. In addition to MAE, which was used as the loss function in the MLP training, Root Mean Square Error (RMSE), Bias, and Index of Agreement (IA) were computed. Compared to MAE, RSME, and Bias, IA is not often reported in experiments; therefore, it requires a more detailed introduction. IA is defined as follows (3):

$$IA = 1 - \left(\frac{SSE}{\sum_{i=1}^{n} \left(|\hat{y}_i - \bar{y}| + |y_i - \bar{y}|\right)^2} \right) \tag{3}$$

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \tag{4}$$

$$\bar{y} = \frac{1}{n} \sum_{i=1}^{n} y_i \tag{5}$$

where y_i and $\hat{y}_i$ are the true and predicted output values for the i-th instance in the data, n is the number of instances. IA is a unitless quantity, which makes it different from MAE and RMSE: IA $\in [0,1]$ — the higher IA, the better [39].

The source code used in this study could be found on: https://github.com/christinabrester/Analytics-project [40].

3. Results

The implemented pipeline was launched for three sites presented in this study. The metrics collected in the outer loop of nested cross-validation were averaged over five-folds and the obtained values were summarized in Tables 4–6 (additionally, Fig. S2 in Supplementary Materials portrays the distribution of these metrics). If we compare the results collected within the Baseline and Alternative scenarios, we can clearly see that in the Alternative scenario, MLP demonstrated the lower performance in terms of MAE, RMSE, and IA for all three sites. Compared to the Baseline scenario, the relative increase in average values of MAE and RMSE in the Alternative scenario equals 33% and 25% for Vuorela, 31% and 13% for Kuopio, and 22% and 12% for Kuopio (Savilahti). The relative decrease in average values of IA does

not exceed 3% for all the sites. This implies that MLP trained on historical weather observations is less effective in predicting solar PV output than MLP trained on historical NWP data, given that both models use NWP data to make predictions for unseen test samples. However, this decrease in the MLP performance is relatively moderate and, depending on the application and requirements, it might be acceptable if no historical NWP data are available for the model training. The same results for LSTM are presented in Supplementary Materials, Tables S1–S3.

Next, the more detailed analysis of model selection in the inner loop showed that the validation MAE was the lowest for MLP configurations with two hidden layers (p^{nn}), activation functions *relu* or *tanh* (p^{act}), and a short "look-back" (p^h), which was relevant for both scenarios, i.e., Baseline and Alternative, and all the sites (Supplementary Materials, Table S4). Also, in the Alternative scenario, bigger networks and more epochs were selected, compared to the Baseline scenario, however, consistency in selecting small values for the "look-back" parameter p^h is again seen for both scenarios. Based on this analysis we may conclude that only several recent hours contain the most relevant data for solar PV forecasting, and the use of more complex models such as LSTM, dealing with long time series data, is not justifiable (the same analysis for LSTM also showed that small values of p^h were selected; the results are summarized in Supplementary Materials, Table S5). This explains why LSTM did not enable us to improve the accuracy achieved with MLP. Our findings are consistent with the results presented by Ref. [17]; where multilayer feed-forward and recurrent ANNs reached similar MAE values in day-ahead solar PV forecasting for individual and aggregated systems.

In addition to the Baseline and Alternative scenarios, we tested the Optimistic scenario, wherein model training and evaluation were performed on historical weather observations, assuming that NWP data would be rather close to them. Tables 4–6 show that in this scenario, generalization error estimates are optimistically low compared to the Baseline and Alternative scenarios because the Optimistic scenario does not consider the error associated with weather forecasts in the input variables. Therefore, if historical NWP data are not available for the experiments, particularly for model evaluation, the use of the Optimistic scenario might be justified but with the important caveat of its positive bias. Moreover, as can be seen from the results, the average values of MAE and RMSE increase drastically (up to several times) when weather observations are substituted for weather forecasts (i.e., Optimistic versus Baseline in Tables 4–6). This implies that by improving the quality of NWP forecasts, it is possible to eliminate a significant part of the prediction error in the Baseline scenario as well as the Alternative scenario.

Another perspective of the current analysis aims to compare ANN with the physical model. As seen in the results, even the Alternative scenario yielded better MAE, RSME, and IA for all three sites than the physical model. In general, while being trained on the collected solar PV data, ANN indirectly accounts for the system properties and local conditions, which can heavily affect the performance of physical forecasts.

Table 4
The model performance evaluation on the test data for the private house in Vuorela. MAE, RMSE, and Bias normalized by the nominal capacity of the PV system are denoted by nMAE, nRMSE, and nBias, respectively.

Metric	Physical model	MLP		
		Baseline	Alternative	Optimistic
MAE, [W]	614.243	197.672	262.269	138.937
nMAE, [W/Wp]	0.07159	0.02304	0.03057	0.01619
RMSE, [W]	1039.332	414.811	520.160	307.518
nRMSE, [W/Wp]	0.12113	0.04835	0.06062	0.03584
Bias, [W],	538.54	19.790	−28.415	−11.513
nBias, [W/Wp]	0.06277	0.00231	−0.00331	−0.00134
IA	0.884	0.972	0.948	0.984

Table 5
The model performance evaluation on the test data for the private house in Kuopio. MAE, RMSE, and Bias normalized by the nominal capacity of the PV system are denoted by nMAE, nRMSE, and nBias, respectively.

Metric	Physical model	MLP		
		Baseline	Alternative	Optimistic
MAE, [W]	344.866	220.470	288.751	69.767
nMAE, [W/Wp]	0.05225	0.03340	0.04375	0.01057
RMSE, [W]	644.593	472.356	536.047	153.188
nRMSE, [W/Wp]	0.09767	0.07157	0.08122	0.02321
Bias, [W],	163.699	25.700	−63.439	12.543
nBias, [W/Wp]	0.02480	0.00389	−0.00961	0.00190
IA	0.943	0.965	0.948	0.996

Table 6
The model performance evaluation on the test data for the institutional building in Kuopio (Savilahti). MAE, RMSE, and Bias normalized by the nominal capacity of the PV system are denoted by nMAE, nRMSE, and nBias, respectively.

Metric	Physical model	MLP		
		Baseline	Alternative	Optimistic
MAE, [W]	878.337	709.003	862.733	172.032
nMAE, [W/Wp]	0.04331	0.03496	0.04254	0.00848
RMSE, [W]	1690.441	1456.099	1634.108	436.507
nRMSE, [W/Wp]	0.08336	0.07180	0.08058	0.02152
Bias, [W],	143.746	8.478	−35.982	60.715
nBias, [W/Wp]	0.00709	0.00042	−0.00177	0.00299
IA	0.960	0.970	0.960	0.997

For instance, in Fig. 3, we can see that for the private house in Vuorela, the physical model tends to overestimate the solar PV output, which is explained by the shading from the nearby trees. For the private house in Kuopio, overestimation by the physical model is less frequent but still takes place. The reason for that is not obvious, and therefore it is not discussed in this study. However, for the institutional building in Kuopio (Savilahti), the physical model is quite a good alternative to ANN.

Lastly, Fig. 4 portrays predictions made by the physical model and MLP in different scenarios for six days in July 2020. In the Optimistic scenario, MLP predicts solar PV output accurately, which implies that errors in the Baseline and Alternative scenarios are associated with inaccuracies in weather forecasting, i.e., in NWP data. Similar visualizations, but interactive and for all days, might be found on GitHub: https://christinabrester.github.io/Analytics-project (in the modeling tab) [40].

4. Discussion

Day-ahead solar PV forecasting assumes that either solar panel specifications are precisely known to apply physical models or historical solar PV output is available to train data-driven models, and in the case of hybrid models, both are needed. It is generally accepted that data-driven models require more data to be set up since they are trained using historical solar PV output and NWP data [41]. However, specifications of solar PV installations (e.g., inclination and orientation) are often unknown or contain mistakes, which makes physical models inadequate [42]. Furthermore, to produce accurate predictions, physics-based models need to account for site-specific shading patterns, which is not a trivial task. In this study, we presented the physical model forecast without applying any site-specific adjustment, which is indeed expected to produce a less accurate output [11]. In contrast, data-driven models indirectly learn local conditions and specifications even though they need more data to train. To mitigate this need for more data, we consider the Alternative scenario, wherein data-driven models, specifically ANNs, are trained using historical weather observations and tested on NWP data. This scenario is supported by the fact that there are several resources from which historical weather observations might be easily

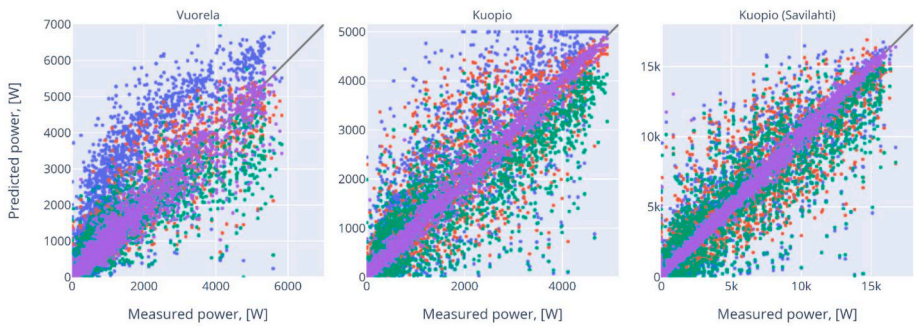

Fig. 3. The scatter plot of measured power versus the corresponding predicted power on the test data.

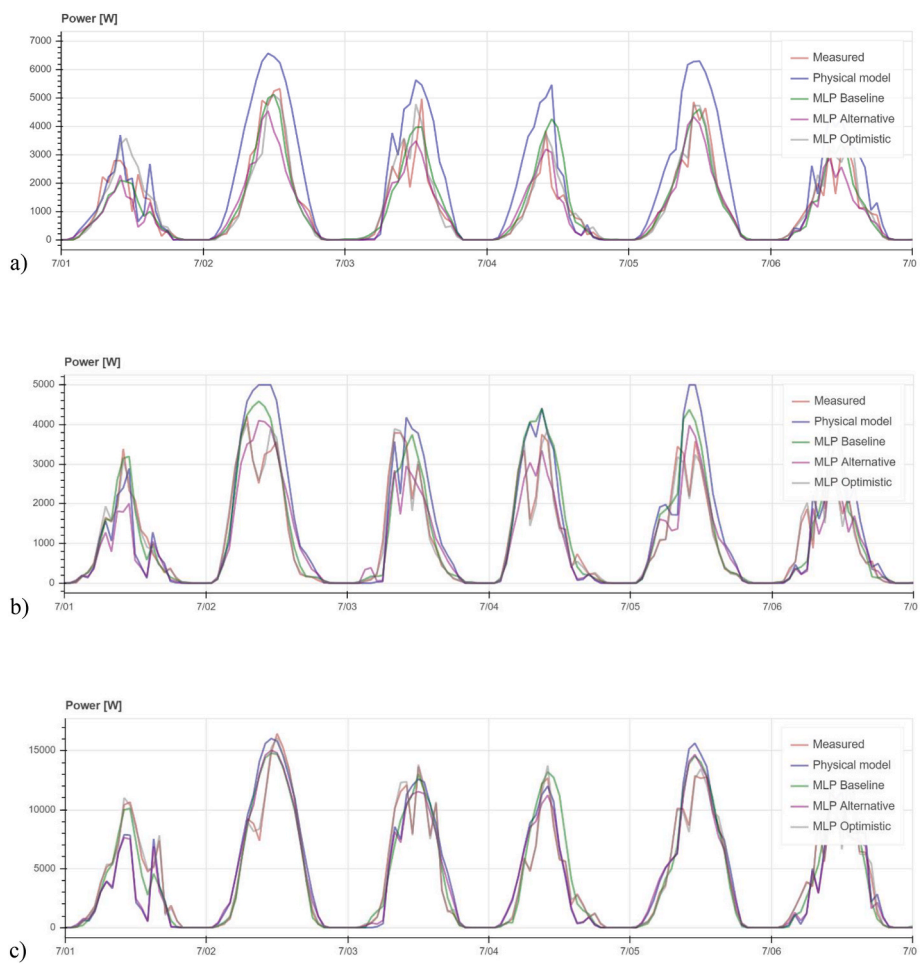

Fig. 4. Outputs of the physical model and MLP on the test data in different scenarios for a) Vuorela, b) Kuopio, and c) Kuopio (Savilahti) for six days in July 2020.

retrieved for training models.

The experimental results obtained for three sites in eastern Finland show that when ANN is trained on historical weather observations (the Alternative scenario), its performance decreases compared to the Baseline scenario. The possible reason for such a decrease is that when trained on NWP data, the predictive model becomes inherently more robust, as the learning algorithm needs to cope with uncertainty in weather forecasts. This makes the model ready for operational use and testing on NWP data, compared to the situation when it is trained on "perfect" observational data. Also, there is always a risk of data drift (changes in distribution) when training and test data come from different sources. All this explains why in many studies data-driven models are trained and tested using the same source of data, i.e., NWP.

Nevertheless, for all three sites, the results of ANN in the Alternative scenario are still better than the results of the physical model. This might be also explained with the relatively high quality of NWP forecasts used in this study: they approximate weather observations accurately enough, therefore, errors in forecasts or changes in distribution when switching between two sources of data are moderate. If historical NWP data are not available for training, the proposed Alternative scenario might be considered instead of the Baseline scenario. Another possible way to deal with the shortage of training data is transfer learning, which is even applicable to new installations, when PV output data have not been collected or do not contain all seasons [43,44]. The main idea is to start with a model trained for another PV installation and then perform fine-tuning with available data. Then, there are also models applied to

irregularly measured weather data, which allow generating gridded weather measurements regularly placed in space. When estimating solar PV output [45], trained ANN in three scenarios (similarly, different sources of training and test data were examined) to show that if in situ weather measurements are not available for a particular place, estimates produced by the Global Data Assimilation System (GDAS) could be used instead. Thus, our work complements other existing studies, in which the shortage of data in solar PV forecasting or estimating is addressed (i.e., historical solar PV data, NWP data in forecasting, or weather observations in estimating).

Additionally, it should be kept in mind that in the Optimistic scenario generalization error estimates for day-ahead solar PV forecasting are optimistically biased because the model accuracy is expected to be lower in operational use when weather forecasting is applied instead of observational data. This drop in accuracy, which might be estimated as a difference between the model performance in the Alternative and Optimistic scenarios in this study, is explained with the uncertainty in NWP forecasts, especially in solar radiation (e.g. Refs. [46–48], which is the main driver of PV output. According to Ref. [49]; the NWP forecast model used in this study produced solar radiation forecasts for up to 30 h, with an average relative bias of around −3%–1% and an average relative RMSE of 28%–36%. To further illustrate the quality of NWP forecasts used in this study, we provide three scatter plots in Supplementary Materials (Fig. S3), portraying observed versus forecast values of temperature, global, and diffuse radiation. Although weather observations were collected by one meteorological station in Savilahti, the scatter plots look quite similar, which allows us to assume that the same weather observations might be used in modeling for all three sites and this does not affect the results significantly. Thus, the uncertainty in NWP forecasts, which is ignored in the Optimistic scenario, makes it the least suitable for evaluating the model performance in day-ahead solar PV forecasting. However, when estimating (not forecasting) solar PV output for different weather conditions, e.g., for filling gaps in historical solar PV output data or for simulating power generation with PV systems, the Optimistic scenario should be preferred to other scenarios.

In this study, we implemented nested cross-validation to avoid model selection bias while estimating ANN performance. As a result, for each scenario, we obtained five different models with their own hyperparameter values and weight coefficients. These models were used to estimate the generalization error. However, in practice, there is often one more step when the final model is trained on the whole data. If several data-driven approaches are considered (e.g., ANN, random forest, and gradient boosted trees), the best one is first selected based on the results of nested cross-validation. Then, the model hyperparameters are defined in the inner loop executed on the whole data; and after that, the final model is trained on the whole data in one run of the outer loop.

5. Conclusion

We investigated the ANN performance in day-ahead solar PV forecasting using different training data, i.e., historical weather observations and historical NWP data. As exemplified with the conventional MLP architecture, ANN training and hyperparameter tuning are slightly less effective with the use of historical weather observations than with historical NWP data. However, the use of weather observations makes model building easier since these data are widely available in online archives. Even after switching to NWP data in operational use, ANN also demonstrates high performance compared to the physical model.

All in all, in day-ahead solar PV forecasting, historical NWP forecasts should be used for training and testing data-driven models, if these forecasts are available. In case of the shortage of historical NWP data, the use of historical weather observations for training forecasting models could be a good alternative to physical models as it has been shown in this study. However, when models are built for estimating solar PV output (not forecasting), weather observations should always be utilized for training and testing purposes.

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CRediT authorship contribution statement

Christina Brester: Methodology, Software, Writing – original draft. **Viivi Kallio-Myers:** Data curation, Writing – review & editing. **Anders V. Lindfors:** Supervision, Data curation, Writing – review & editing. **Mikko Kolehmainen:** Supervision, Writing – review & editing. **Harri Niska:** Funding acquisition, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.renene.2023.02.130.

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